

Is minimal (non-custodial) Randall–Sundrum still better than the Standard Model on *flavor*?

A confirm-or-kill reassessment of the anarchic warped-flavor claim

5D-Neutrino-Mixing project — quark-sector reassessment note

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Abstract

We ask, narrowly and quantitatively, whether minimal anarchic Randall–Sundrum (RS) makes *any* flavor statement — at an indirectly reachable M_{KK} — that the Standard Model (SM) does not already make, and that is not already collapsed to “ $\sim$ SM.” Working entirely from a 9.9×10^5 -draw anarchic forward ensemble (the literature “strawman” lane), with masses and tree-level CKM *fixed* and all flavor/CP/electroweak observables held out as a test set, we build an *SM-overlap scoreboard*: for every NP-sensitive quark channel we compute the M_{KK} below which a $> 1\sigma$ deviation from the SM is *generic*, and overlay the scales the model must survive (ε_K and the oblique S, T, U). The result is unambiguous. Every flavor deviation has fallen below 1σ by $M_{\text{KK}} \approx 1.9$ TeV, while survival requires $M_{\text{KK}} \gtrsim 19$ TeV from ε_K and $\gtrsim 18$ – 20 TeV irreducibly from oblique T — a full decade of separation. Custodial symmetry relieves the electroweak floor ($\sim 16 \rightarrow 6$ TeV in our oblique proxy) but leaves the ε_K wall untouched, because ε_K is a custodial-invariant KK-gluon operator. We conclude (B): *minimal anarchic RS retains exactly one genuine virtue absent in the SM — the geometric, single-input origin of both the mass hierarchy and the RS-GIM FCNC-suppression pattern — but as a predictive flavor theory it no longer buys anything; every would-be prediction is unobservable at any viable scale, and viability itself must be purchased with custodial protection and explicit flavor alignment, structure the SM never required.* Throughout, every floor is tagged with its model lane; the numbers here are the **anarchic** (Lane A) floors by design.

1 The question and a fair standard

The interesting comparison is *not* whether RS can fit the fermion masses and the CKM matrix. In the bulk-fermion construction that is the premise: exponentially different zero-mode overlaps, set by one $O(1)$ bulk-mass parameter c per field, turn anarchic $O(1)$ five-dimensional Yukawas into the observed 4D hierarchy. The real question is whether, *after* fitting masses and mixings, the same geometry yields a distinctive, still-viable pattern of flavor and CP violation that the SM does not already describe at least as well.

We therefore impose an explicit **fit-set / test-set split**. The fit set contains only what RS is allowed to claim geometrically: quark masses and the tree-level CKM angles and phase. The test set is everything else — neutral-meson mixing ($\varepsilon_K, \Delta m_K, \Delta m_{B_{d,s}}$, the B mixing phases), D^0 mixing, and electroweak precision ($S, T, U, Z \rightarrow b\bar{b}$). This is also what makes the comparison fair to the SM: the SM benchmark uses tree-level CKM inputs only, so no possible new physics in the mixing observables is fed back into the reference fit used to test those same observables. “RS does better” then means a *predictive surplus* on the test set after the fit set is fixed — not merely extra flexibility.

Two benchmarking points must be stated up front. First, the anarchic ε_K wall (Sec. 4) is the baseline the literature *criticizes*; it is not “our” production model. Second, RS does **not** reduce the number of physical flavor parameters relative to the SM: the quark sector still carries

two complex 5D Yukawa matrices plus bulk-mass matrices [1]. The claim was always *structure from anarchy plus localization*, never fewer knobs. RS must therefore win, if at all, on structure, prior-naturalness, and out-of-sample predictions.

Space of honest conclusions. We pre-registered three outcomes: (A) RS wins narrowly — some generic anarchic correlation, distinct from the SM at $> 1\sigma$ of current resolution, realized at an M_{KK} that simultaneously survives ε_K and S, T, U ; (B) RS is reduced to a mass-hierarchy mechanism plus a non-falsifiable FCNC correlation; (C) RS flavor adds nothing whatever. We commit to landing wherever the data point, by *trying to rescue (A) and documenting the failure*. We explicitly decline (C): the geometric-hierarchy virtue is real and a referee would correctly defend it. The spine of the paper is one question: *is there any M_{KK} at which a generic anarchic point both survives and still shows a measurable deviation?*

2 What RS genuinely buys — and what it does not

The one SM-absent virtue is worth steelmanning. The same wavefunction overlap $f(c)$ that produces the five-decade mass hierarchy also sets the flavor-changing KK couplings, with the celebrated RS-GIM scaling

$$(\delta_{\text{FCNC}})_{ij} \sim \frac{\sqrt{m_i m_j}}{v}, \quad (1)$$

so that light-fermion FCNCs are automatically suppressed by the *same* input that makes those fermions light. A single geometric structure generates two phenomena the SM merely parametrizes (the hierarchy) or assumes (GIM). That is a genuine explanatory economy, and it is the strongest thing RS has to say about flavor.

What RS does *not* buy is equally important and is the SM’s side of the ledger. The SM already accounts for *all* observed flavor and CP violation as a precision null test: once masses and tree-level CKM are fixed, the loop-level mixing observables are reproduced with *no added structure*, no new parameters, and no tuning. RS, to remain viable, must add custodial protection (for T , Sec. 8) and explicit flavor alignment (for ε_K), and even then does not reduce the parameter count. So the contest is asymmetric in a way the rest of this note quantifies: RS offers a *story* the SM lacks, but on *predictions* the SM is at least as good and structurally cheaper.

3 The recentered electroweak landscape (2008 $\rightarrow$ 2025)

A reassessment is timely now, and was not in 2008, for three reasons. The Higgs has been found, moving RS’s *raison d’être* from “trigger EWSB” to “flavor + composite Higgs” and re-introducing the little hierarchy v/f . The ε_K SM tension sharpened with 2025 lattice inputs. And — the cleanest lever — the oblique ellipse **recentered on the SM**. Figure 1 overlays the 2008 fit (CGHNP, $S = 0.07 \pm 0.10$, $T = 0.16 \pm 0.10$) against PDG-2025 ($S = 0.026 \pm 0.075$, $T = 0.047 \pm 0.066$, $\rho = 0.90$; and the U -free variant). The 2008 ellipse sat at *positive* T , exactly where a minimal RS KK tower wants to live; the 2025 ellipse sits on the SM point, so the same RS T -wedge is now disfavored.

Quantitatively, the recentering *raises* the minimal floor and custodial *lowers* it (Table 1). Against PDG-2025 (U -fixed) the minimal oblique proxy is excluded up to 16.0 TeV, versus 10.4 TeV against the looser 2008 ellipse — the tighter, recentered fit is a real lever. Custodial P_{LR} protection drops the same floor to 5.8 TeV (the residual is set by S , which custodial does not protect). These are oblique-only proxy floors at fixed $(v, \sin^2 \theta_W, L=35)$; the lane-independent S, T, U *existence* floor from the full scan is 18–20 TeV (Sec. 5).

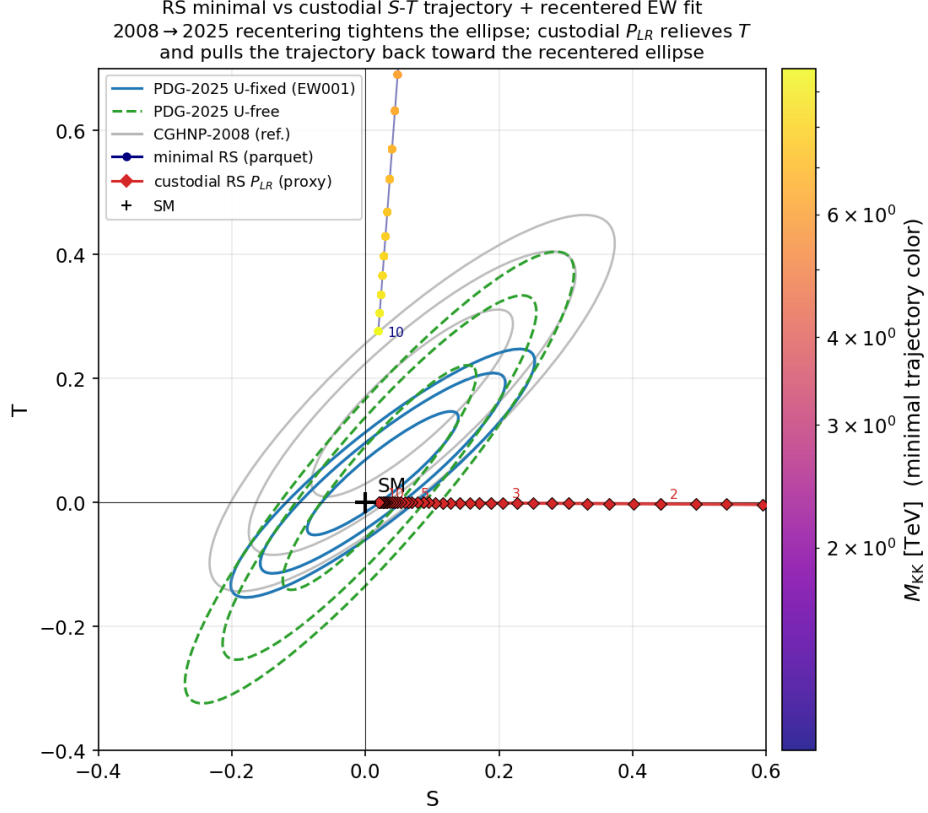


Figure 1: Recentered S - T plane. The minimal RS proxy (navy, colored by M_{KK}) climbs steeply in T , leaving every ellipse; the custodial P_{LR} proxy (red) hugs $T \approx 0$ along the S axis. The 2008 $\rightarrow$ 2025 recentering (grey $\rightarrow$ blue/green ellipses) tightens the fit *and* moves it onto the SM, raising the minimal oblique floor (Table 1).

4 The flavor wall: anarchy over-saturates ε_K

The strongest generic bound on anarchic RS comes from the kaon system, where the left-right $\Delta S = 2$ operator C_4 is chirally and RG-enhanced. Figure 2 shows the anarchic $|\varepsilon_K|$ cloud with 5/50/95% quantiles. The 95% quantile crosses the experimental window only near $M_{KK} \sim 10$ TeV (paper-era inputs); with current inputs the *median* floor is higher. Figure 5 (right) makes the mechanism explicit: the KK contribution to the kaon M_{12} is $\sim 10^2 \times$ more important in its imaginary (CP-violating) part than its real part, the hallmark of the C_4 -dominated ε_K problem.

The wall is robust to the dominant systematic, $|V_{cb}|$, which controls $\varepsilon_K^{\text{SM}}$ and hence the NP budget. Rescaling the per-draw ε_K exactly across the $|V_{cb}|$ range gives an anarchic median floor of

$$\varepsilon_K \text{ floor} \approx 9\text{--}24 \text{ TeV}$$

(inclusive vs. exclusive $|V_{cb}|$, realized-Im vs. modulus phase convention). Every value is an order of magnitude above the flavor-*deviation* floor of Sec. 6, so the verdict does not depend on this choice; the much-quoted 5σ ε_K tension is corroborating, not load-bearing.

5 The irreducible wall: oblique T and the existence floor

Unlike flavor, which can in principle be tuned or aligned away, the oblique T parameter has *no* Yukawa freedom: it is generated by the KK tower’s electroweak mixing and is essentially the same for every anarchic draw. Figure 3 separates the *existence* floor (best/tuned point,

Table 1: C1 oblique floors: largest M_{KK} [TeV] excluded at 95% by the S, T proxy, for each EW fit. Recentering raises the minimal floor; custodial relieves it at every fit but cannot reach the few-TeV regime because S is unprotected.

EW model	CGHNP-2008	PDG-2025 (U -fixed)	PDG-2025 (U -free)
minimal RS	10.4	16.0	13.2
custodial P_{LR}	7.6	5.8	4.6

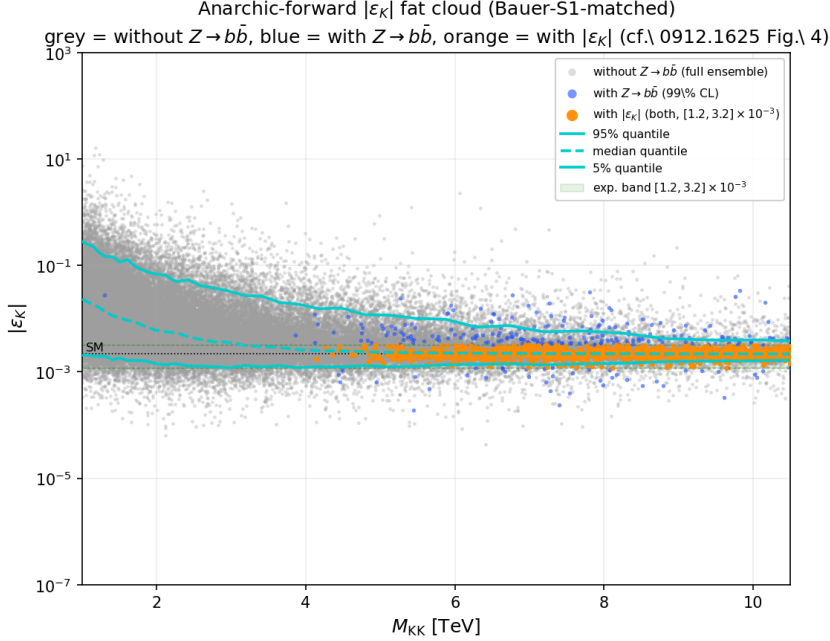


Figure 2: Anarchic (Lane A) $|\varepsilon_K|$ cloud vs M_{KK} : grey = mass+CKM consistent (no $Z \rightarrow b\bar{b}$ cut), blue = also $Z \rightarrow b\bar{b}$ consistent, orange = also in the ε_K window. The 95% quantile reaches the window edge near 10 TeV (paper-era); the typical wall is higher with current inputs.

per- M_{KK} minimum) from the *typical* floor (median). For the flavor observables the existence floor crashes below 1 TeV — a tuned point can always evade any single flavor constraint — but the *combined* existence floor sits at 18–20 TeV, set by S, T, U , which the minimum tracks the median because there is nothing to tune. This is the lane-independent wall: it stands in every flavor scenario, anarchic or aligned.

6 The decisive test: the SM-overlap scoreboard

We now run the (A)-vs-(B) discriminator directly. From the co-registered 9.9×10^5 -draw ensemble of complex NP M_{12} values for $K/B_d/B_s/D$ (gated to the 1.1×10^5 mass+CKM-consistent draws), we compute, per channel and per M_{KK} tile, the generic (median) deviation from the SM in units of the current experimental resolution. Crucially, the B -meson *phases* are computed KI-correctly, from the stored complex M_{12} against a *complex* SM box,

$$\phi_q^\Delta = \arg(1 + M_{12}^{\text{NP}}/M_{12}^{\text{SM-box}}), \quad M_{12}^{\text{SM-box}} \text{ complex}, \quad (2)$$

not a real- M_{12}^{SM} adapter (which is not rephasing-invariant). The channels and their 1σ sensitivities are ϕ_s (16 mrad), C_{B_s} (0.09), $\sin 2\beta$ (0.017), C_{B_d} (0.08), and D^0 amplitude (vs. Δm_D^{exp} , long-distance-dominated and hence a conservative bound, not a hard exclusion).

The result (Fig. 4, Table 2) is decisive. The most NP-sensitive channel, the B_d mixing phase, has a generic deviation that crosses below 1σ at $M_{KK} = 1.9$ TeV; the others cross even lower,

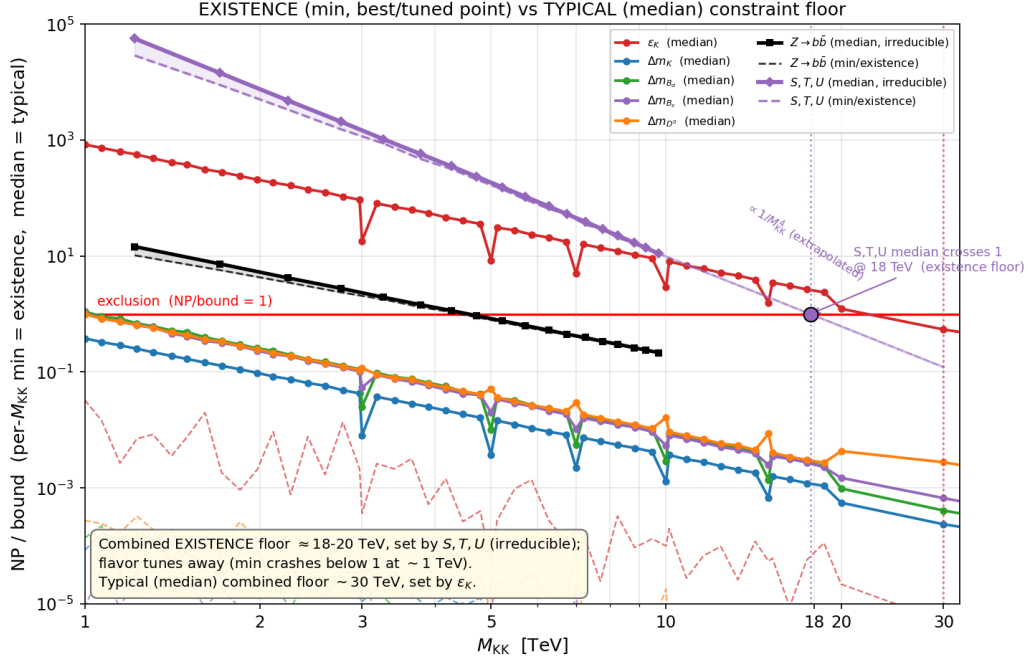


Figure 3: Existence (min, best point) vs typical (median) floors. Flavor tunes away (its min crashes below 1 TeV); S, T, U does not (min $\approx$ median), setting an irreducible existence floor of 18–20 TeV. The typical combined floor (~ 30 TeV here) is set by ϵ_K .

and C_{B_s} and D^0 never reach 1σ even at 1 TeV. Meanwhile survival requires $M_{KK} \gtrsim 19$ TeV (ϵ_K) and $\gtrsim 18$ –20 TeV (S, T, U). At the survival floor every channel sits at 10^{-2} – $10^{-3}\sigma$ — utterly SM-like. The robustness is double: even the 95% tail (the most NP-active draws) is sub- 1σ for $M_{KK} \geq 18.95$ TeV (the B_d phase tail crosses 1σ at 18.5 TeV, just inside the S, T, U band, and is 0.71σ by 20 TeV). There is no M_{KK} at which a surviving anarchic point shows an observable flavor deviation. **Conclusion (A) is dead; (B) is confirmed.**

7 Electroweak side notes

Two corrections keep the diagnosis honest. First, $Z \rightarrow b\bar{b}$ is *not* the driver: a corrected treatment puts its floor near 5 TeV (Fig. 5), well below ϵ_K and S, T, U , so the binding constraints are genuinely ϵ_K and T , not $Z \rightarrow b\bar{b}$. Second, the kaon Im/Re asymmetry of Fig. 5 (right, $\sim 10^2$) is the fingerprint of the C_4 operator and confirms our matching against the published anarchic results.

8 The cures and what they cost

What would it take to rescue a low M_{KK} ? Two distinct additions, neither for free.

Custodial protection fixes the electroweak sector, not ϵ_K . The custodial $SU(2)_L \times SU(2)_R \times U(1)_X \times P_{LR}$ structure cancels the leading T ($SU(2)_R$ [5]) and protects the $Z b_L \bar{b}_L$ vertex (the discrete P_{LR} [6]), relieving the oblique floor from ~ 16 TeV to ~ 6 TeV (Table 1); the residual is set by S , which custodial does *not* protect. But ϵ_K is a color-octet KK-gluon left–right (C_4) operator, a *singlet* under the electroweak custodial group, and is therefore untouched by custodialization. This is not an inference from operator counting alone: Blanke *et al.* [4] computed the full $\Delta F = 2$ set *inside the custodial model* and still find $M_{KK} \gtrsim 18$ TeV (fine-tuning < 20) / $\gtrsim 30$ TeV (< 10); Csáki–Falkowski–Weiler [8] make the split explicit — custodial

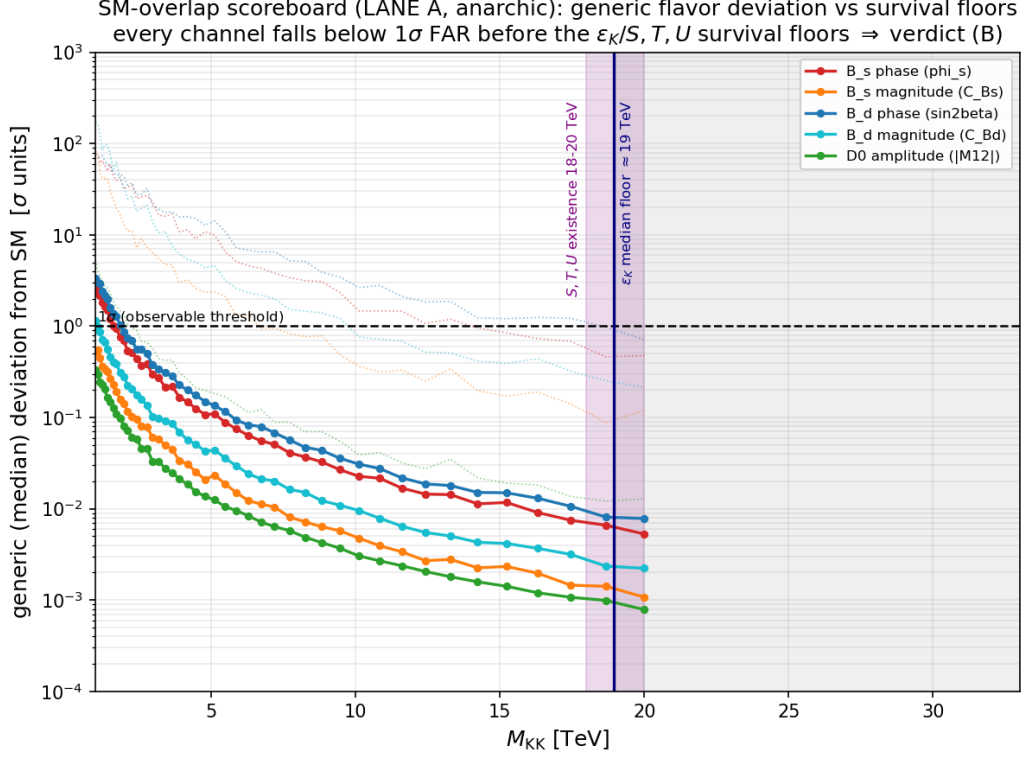


Figure 4: The decisive figure. Generic (median) deviation from the SM, in σ units, vs M_{KK} , for every NP-sensitive quark channel (dotted = 95% tail). The dashed line is the 1σ observability threshold; the shaded bands are the ε_K median and S, T, U existence survival floors. Every channel falls below 1σ by $M_{KK} \approx 1.9$ TeV — a full decade below the survival floors. At the survival floor all channels sit at 10^{-2} – $10^{-3}\sigma$.

passes electroweak precision at ~ 3 TeV while ε_K forces $M_{KK} \gtrsim 21$ – 33 TeV. Figure 6 collects this: the electroweak floors (S, T, U and $Z \rightarrow b\bar{b}$) move to a few TeV under custodial, while the ε_K wall does not move at all, in our numbers and in the literature alike. Custodial buys electroweak viability, and nothing on flavor.

Flavor alignment fixes ε_K , by adding structure. Aligning or degenerating the flavor sector — 5D MFV, a horizontal $U(1)$, or a $U(3)$ degeneracy of the right-handed down quarks (Bauer’s “S2”) — can drive the ε_K floor from ~ 19 TeV (anarchic, S1) down toward ~ 1.8 TeV (Fig. 7). But this trades “anarchy *predicts*” for “symmetry *imposes*”: the cloud stays anarchically fat, it simply gains far more ε_K -consistent points once the right-handed down mixing that feeds C_4 is removed by hand. A down-aligned cure must additionally pay the D^0 /charm cross-check, since relieving the kaon can shift residual flavor violation into the up sector (D^0 is conservative here, being long-distance-dominated). The net flavor structure is then no more predictive, and no more economical, than the SM’s.

9 Verdict

We land on (B). Minimal anarchic RS retains exactly one virtue the SM lacks: a single geometric input generates both the fermion-mass hierarchy and the RS-GIM FCNC-suppression pattern. That virtue is real, and we do not dispute it. But as a *flavor theory* it no longer buys a viable TeV-scale prediction. Run forward with anarchy, the same input forces $M_{KK} \gtrsim 9$ – 24 TeV from ε_K and $\gtrsim 18$ – 20 TeV irreducibly from oblique T , while *every* would-be flavor prediction — ϕ_s , $S_{\psi\phi}$, C_{B_d} , C_{B_s} , the D^0 phase — has collapsed onto the SM point by ~ 2 TeV, a decade below

Table 2: SM-overlap scoreboard: M_{KK} [TeV] below which a $> 1\sigma$ deviation is *generic* (median draw), per channel, vs the survival floors. The largest deviation floor (1.9 TeV) is a decade below the binding survival floor.

Channel (quark)	generic $> 1\sigma$ deviation floor [TeV]
B_d phase ($\sin 2\beta$)	1.87
B_s phase (ϕ_s)	1.64
B_d magnitude (C_{B_d})	1.06
B_s magnitude (C_{B_s})	never reaches 1σ
D^0 amplitude	never reaches 1σ
<i>survival</i> : ε_K median	~ 19 (band 9–24)
<i>survival</i> : S, T, U existence	18–20

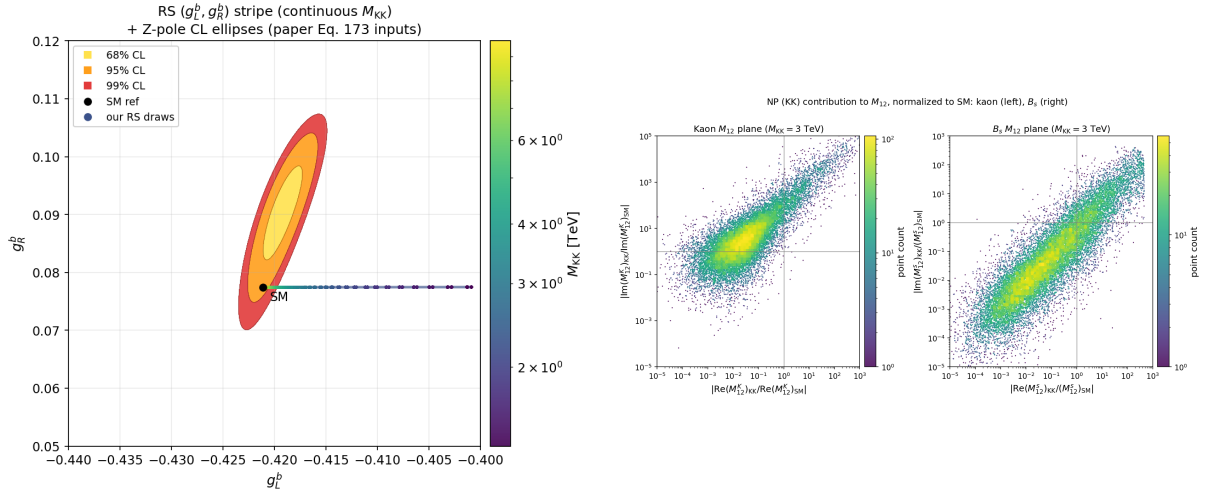


Figure 5: Left: (g_L^b, g_R^b) stripe with the Z -pole CL ellipses — the $Z \rightarrow b\bar{b}$ floor is ~ 5 TeV, not a driver. Right: the kaon (and B_s) M_{12} planes, showing the $\sim 10^2$ imaginary-over-real KK enhancement that drives ε_K .

the survival floor. The recentering of the S – T ellipse onto the SM disfavors the RS T -wedge that 2008 data tolerated; custodial symmetry rescues the electroweak sector but not ε_K ; and alignment rescues ε_K only by importing the flavor structure that anarchic RS was supposed to make unnecessary.

The honest bottom line is symmetric. **What RS does that the SM does not:** it tells a geometric story for why the hierarchy and GIM look the way they do. **What the SM does better:** it accounts for all observed flavor and CP violation as a loop-level null test, with no added structure, no parameter reduction conceded by RS, and none of the custodial-plus-alignment scaffolding RS needs merely to survive. The RS-GIM correlation is real but non-falsifiable at any reachable scale; on the predictions that could distinguish the two theories, the SM is at least as good and structurally cheaper. Minimal warped flavor today is a mechanism for the mass hierarchy — no longer a competitive theory of flavor.

Lane discipline. All floors quoted here are the **anarchic** (Lane A) floors — the literature strawman the note interrogates — not the project’s simplified-MFV production lane (~ 7 TeV, no V_{5KM}) or the fully aligned FPR ideal (~ 2 TeV). The 7→2 TeV gap in the production program is exactly the principled alignment (Sec. 8) not yet wired in; it does not affect the anarchic verdict above.

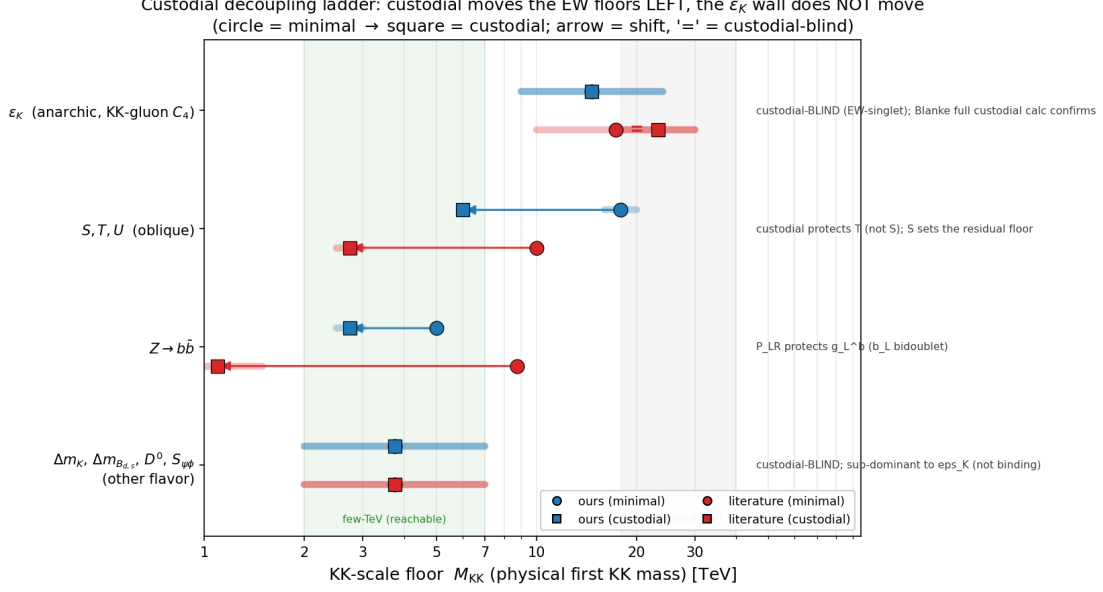


Figure 6: Custodial decoupling ladder. Per-constraint KK-scale floor, minimal (circle) $\rightarrow$ custodial (square), ours (blue) vs literature (red). The electroweak constraints (S, T, U , $Z \rightarrow b\bar{b}$) shift *left* under custodialization (arrows); ϵ_K and the other flavor constraints do not move (“=”), because the binding kaon operator is a color-octet KK-gluon, blind to the electroweak custodial group. Literature anchors: [5, 7, 8, 4, 3]. (Leading custodial physics only; the subleading custodial flavor effects — EW KK-boson $\Delta F = 2$, custodian loops — are not in our proxy and do not change the ϵ_K wall.)

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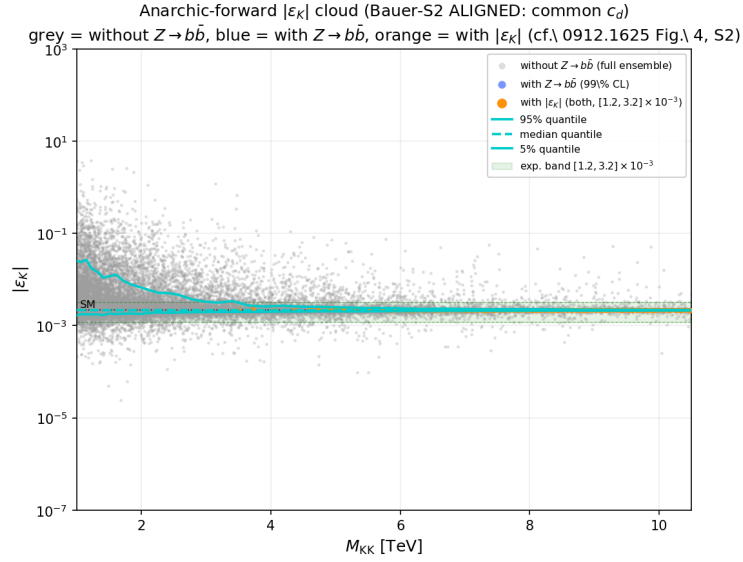


Figure 7: Alignment endpoint (Bauer-S2, common right-handed down c_d via a $U(3)$ degeneracy): the cloud stays fat but gains ε_K -consistent points, dropping the floor toward ~ 1.8 TeV — a cure that imposes structure rather than predicting it.